

Degree Project in the Field of Technology Media Technology and the Main Field of Study Computer Science and Engineering

Second cycle, 30 credits

Analyzing User Engagement to Fact-check on Personal Messaging Apps

A Case Study on Cek Fakta Tempo's WhatsApp Chatbot Tip Line

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Master’s Programme, Media Management, 120 credits
Date: March 23, 2026
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Examiner: Tino Weinkauf
School of Electrical Engineering and Computer Science
Swedish title: En analys av användarengagemang i faktagranskning via privata meddelandeplattformar
Swedish subtitle: En fallstudie av Cek Fakta Tempos WhatsApp-chatbot

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ABSTRACT

The proliferation of misinformation on encrypted personal messaging platforms, particularly WhatsApp, presents considerable challenges for content moderation and information verification. While existing research in communication and media studies has predominantly investigated misinformation through sociocultural or behavioral lenses, fewer studies have explored how design and technical affordances influence user engagement with fact-checking tools within closed digital environments. This thesis addresses this gap by examining the user experience and interaction design of Cek Fakta Tempo's WhatsApp chatbot, a misinformation tip line developed by Indonesia's Tempo media group in collaboration with Meedan. Utilizing User Engagement Theory and evaluated through the User Engagement Scale–Short Form, this study employs a mixed-methods approach integrating system usage metrics, survey-based engagement assessment, and qualitative insights derived from semi-structured interviews. The findings indicate that usability, efficiency, and interface familiarity are primary determinants of engagement, with trust in the news organization fostering sustained use. However, limitations in personalization and information retrieval restrict long-term engagement. The study proposes that enhancing engagement quality depends on addressing these algorithmic frictions and better aligning interface aesthetics with localized cultural communication norms.

SAMMANFATTNING

Spridningen av desinformation på krypterade meddelandeplattformar, särskilt WhatsApp, innebär betydande utmaningar för moderering av innehåll och verifiering av information. Tidigare forskning inom kommunikations- och medievetenskap har huvudsakligen undersökt desinformation ur sociokulturella eller beteendemässiga perspektiv, medan färre studier har

analyserat hur design och tekniska förutsättningar påverkar användarengagemang i faktagranskningsverktyg inom slutna digitala miljöer. Denna avhandling adresserar detta forskningsgap genom att undersöka användarupplevelse och interaktionsdesign i Cek Fakta Tempos WhatsApp-chatbot, en faktagranskningskanal utvecklad av det Indonesiska medieföretaget Tempo i samarbete med Meedan. Studien utgår från teorin om användarengagemang och utvärderas med hjälp av User Engagement Scale–Short Form. En blandad metod används, där systemanvändningsdata, enkätbaserade mätningar av engagemang och kvalitativa insikter från semistrukturerade intervjuer integreras. Resultaten visar att användbarhet, effektivitet och gränssnittets igenkänning är de främsta faktorerna som påverkar engagemang, medan förtroende för nyhetsorganisationen bidrar till långsiktig användning. Begränsningar i personalisering och informationssökning minskar dock det långvariga engagemanget. Studien föreslår att en förbättrad engagemangskvalitet är beroende av att man adresserar dessa algoritmiska friktioner och bättre anpassar gränssnittets estetik till lokaliserade kulturella kommunikationsnormer.

CCS Concepts: • **Human-centered computing** ⟶ **Empirical Studies in HCI; Interactive systems and tools** • **Computing Methodologies** ⟶ **Discourse, dialogue and pragmatics** • **Applied Computing** ⟶ **Sociology.**

Keywords: User Engagement, Chatbot Interaction, Human-Computer Interaction (HCI), Fact-Checking, Misinformation, Private Messaging Platforms

Nyckelord: Användarengagemang, Chatbot-interaktion, Människa-datorinteraktion (HCI), Faktagranskning, Desinformation, Privata meddelandeplattformar

1 INTRODUCTION

Fake news or misinformation is commonly found on the three most popular platforms: Twitter, Facebook and WhatsApp [24]. Due to the limited number of fact-checkers in Southeast Asia, verifying all content generated by internet users becomes a challenging task to complete [50] as Indonesia has the highest number of internet and smartphone users in the region according to Statista[1].

In line with these findings, Indonesian activists and media organizations have been focusing on social media to combat misinformation since 2014's Presidential Election [16]. Additionally, Reuters Institute for the Study of Journalism's Digital News Report 2021 [13] shows that the people of Global South, including Indonesia, considers WhatsApp a medium for spreading misinformation.

[1] https://datareportal.com/reports/digital-2020-indonesia

Unlike the largely open forums of Facebook and Twitter, the closed nature of messaging services complicates the already difficult task of fighting misinformation [20]. For example, during the 2018 elections in Brazil and the 2019 elections in India, it was observed that misinformation was largely shared on WhatsApp public groups even after being verified by popular fact-checking agencies [40].

While fact verification on personal messaging apps is more challenging than social media, study findings suggest tip lines on WhatsApp can be an effective source for identifying potentially misleading content. People are more likely to share debunked information on WhatsApp if it comes from close social connections within their social group, highlighting the importance of both strong ties and in-group members in spreading misinformation [22].

Thus in many countries and regions, Meta–formerly named Facebook, Inc., and The Facebook, Inc–works with independent third-party fact-checking organizations certified through the non-partisan International Fact-Checking Network (IFCN) in launching its WhatsApp third-party fact-checkers program[2]. Out of Meta's 18 partners worldwide, three are based in Indonesia[3], taking part in the program by integrating chatbot technology to conduct fact-checking. Since 2021, Cek Fakta Tempo (Tempo Fact Check–in English) has been licensed to operate Check, a flagship platform scaled up by Meedan, to verify and run fact-checking tip lines on the WhatsApp Business API[4].

While scholarly attention has increasingly focused on the efficacy of fact-checking interventions, a significant research gap persists regarding user engagement with automated chatbots within the "closed" messaging ecosystems of Southeast Asia, such as WhatsApp. Current literature on user engagement often distinguishes between open platforms (e.g., X/Twitter or Facebook), where engagement is measured through public metrics like shares and likes, and closed environments, which necessitate more granular, interaction-based metrics to capture user behavior. Despite the implementation of three fact-checking tip lines and a regional initiative in Indonesia, the interactional dynamics and user perceptions of these private interfaces remain under-theorized.

By conducting a comprehensive analysis, this study attempts to identify the factors contributing to the user engagement on the closed-network tip line by answering the following research questions:

[2] https://transparency.fb.com/en-gb/features/approach-to-ranking/content-distribution-guidelines/misinformation
[3] https://meedan.com/programs/whatsapp-3pfc
[4] https://faq.whatsapp.com/5059120540855664

- **RQ1:** *What are the main incentives for fact-check users to engage with a WhatsApp chatbot tip line?*
- **RQ2:** *What are the affordances and challenges that influence user engagement of a fact-checking chatbot on WhatsApp according to the Short Form of the User Engagement Scale (UES)?*

The focus of this research is not on examining users' perceptions of fact-checking as a process but rather on identifying the factors that enhance engagement with chatbot technology. In other words, the findings aim to contribute to the broader efforts in combating misinformation in closed messaging environments, equipping fact-checking organizations like Tempo with strategies to strengthen audience engagement in a rapidly evolving digital media landscape.

2 BACKGROUND

The following section covers the works related to misinformation on personal messaging apps, the role of misinformation tip line, and user engagement and measuring user engagement of the chatbot tip line.

2.1 The Rise of Misinformation in Personal Messaging Apps

With the increasing global reliance on personal messaging apps, platforms such as WhatsApp and Telegram have become major channels for information exchange. Due to their private, encrypted nature, these platforms present unique challenges and opportunities in the context of misinformation. Unlike open social media platforms like Facebook and Twitter, content in personal messaging apps is often shielded from public scrutiny and moderation efforts, making them an attractive venue for the rapid spread of unverified or misleading information [2]. Recent studies have demonstrated that personal messaging apps, particularly WhatsApp, play a disproportionate role in the dissemination of misinformation, especially in developing countries [7, 14, 17, 39].

Social media-related research emphasizes how algorithmic amplification contributes to the dissemination of harmful, inaccurate, or low-quality material [30]. In personal messaging applications, the amplification is driven by individuals rather than algorithms.

The inherent structure of personal messaging apps, characterized by end-to-end encryption and the presence of smaller, private groups, allows users to share content freely without moderation [40]. WhatsApp's combination of viral content sharing and encrypted personal messaging enables the rapid spread of misinformation [41].

While these design choices enhance privacy, they also complicate efforts to curb misinformation. Misinformation tends to spread rapidly in private networks, as users are often more likely to trust and accept information from close contacts than from public sources [9]. This creates a highly fertile environment for "information cascades" [8] where users unknowingly perpetuate misinformation due to limited fact-checking options and social verification biases.

This explains why WhatsApp's primary efforts to combat misinformation generate barriers to sharing and provide indicators for users to identify and maybe question "viral" material or dubious groups (i.e., groups they are joined to by non-contacts) [42].

2.2 The Role of Misinformation Tip Lines

Fact-checking organizations have emerged as key players in addressing the growing challenge of misinformation in the digital age. One strategy that has gained traction is the implementation of misinformation tip lines [15], which allow users to directly submit potentially false or misleading content for verification. In the context of personal messaging apps, such tip lines can be integrated directly into the user experience through chatbot technology, making it easier for users to quickly report and verify information.

Meedan assists fact-checking organizations in setting up and operating misinformation tip lines using their open-source software, Check Bot [23]. Launched in 2020, this software is a customized WhatsApp business tool to provide a direct channel for users to submit content for verification.

Tip lines on personal messaging apps have been implemented in several fact-checking initiatives and organizations, for example in India, Brazil, and Spain. A tip line that was run during the 2019 Indian elections, received over 37,000 unique messages. In Spain, a fact-checking organization Maldita.es used their tip line and chatbot to analyze over 8,000 fact-checking requests, which allowed them to identify trends in misinformation [17].

In Southeast Asia, the growth of social media and messaging platforms has dramatically shifted how information is disseminated, often at a rate that outpaces regulatory and educational efforts to promote digital literacy [37]. Countries in this region, such as Indonesia, the Philippines, and Malaysia, have seen a surge in misinformation, driven by socio-political factors, language diversity, and high levels of digital engagement [21]. Fact-checking initiatives have emerged in response, as both government and civil society organizations work to combat misinformation and provide the public with accurate information.

The 2014 Presidential Election, the 2017 Jakarta governatorial election, and 2019 presidential election, marked fact-checking urgency where misinformation deeply penetrated political discourse in Indonesia [46]. In response, several Indonesian media outlets—including Tempo [19]—paved their ways to provide fact-checking services and certified by The International Fact-Checking Network (IFCN) supported by partnerships with Cekfakta.com[5], a collaborative fact-checking project initiated by AJI (Alliance of Independent Journalists), MAFINDO (Indonesian Anti-Defamation Society)[6], and AMSI (Indonesian Cyber Media Association).

The Check Bot offers several key features that make it highly adaptable and effective in fast-paced fact-checking environments. First, it allows for fully customizable interaction scenarios, enabling fact-checking teams to configure interactive menus with frequently requested content or critical information as events unfold. Second, the bot facilitates user engagement by allowing audiences to submit tips or queries that may not be addressed by the pre-configured responses.

Third, Check Bot integrates directly with a team's Check database, enabling seamless access to previously fact-checked reports and allowing users to receive verified information on demand. By combining artificial intelligence with interactive menus and a database of verified contents, Check Bot exemplifies the potential of automated tools to support human-led fact-checking efforts on platforms where misinformation is prevalent but difficult to monitor.
Cek Fakta Tempo, launched its misinformation tip line on WhatsApp in November 2021, after the Covid-19 pandemic. This Check Bot helps Cek Fakta Tempo to combat misinformation by providing a way for users to interact with the fact-checking team directly, to report, and to verify information, as well as to identify trends in misinformation from the content submitted to the tip line.

2.3 User Engagement and Measuring User Engagement of Tip Line

User engagement is a quality of user experience characterized by attributes such as challenge, positive affect, interactivity, and perceived user control during interactions with technology [32]. It involves a dynamic process that includes stages of engagement, sustained engagement, disengagement, and reengagement, influenced by both the user and the system [33].

In digital contexts, user engagement is often seen as a multi-dimensional construct that includes Cognitive Engagement, the extent to which users are mentally invested in the content; Emotional Engagement, the feelings that users associate with their interactions [6]; and

[5] https://cekfakta.com/about
[6] https://mafindo.or.id/tentang-mafindo/

Behavioral Engagement, the observable actions taken by users, such as comments, shares, or time spent on a platform [28]. Engagement patterns may differ substantially between chatbots and other digital tools, such as news applications or social media platforms, given their distinct modes of interaction and perceived credibility. Newsreaders, like other digital users, are drawn to a news website's organization, aesthetics, and headlines, but their engagement depends on usability, content relevance, and ease of navigation, with frustration arising from poor design or distracting features [36].

While in marketing, a chatbot's responsiveness and conversational tone directly enhance customer satisfaction, indirectly boosting social media engagement, price premium, and purchase intentions, with satisfaction further driving engagement and behavioral outcomes like higher spending and purchase likelihood [51].

Measuring user engagement is crucial for evaluating the effectiveness of fact-checking tools. High levels of user engagement can indicate that a tool is successfully attracting attention and encouraging critical thinking among users regarding misinformation [28]. For instance, engaged users are more likely to share fact-checked information within their networks [35]. Measurement can inform developers about user preferences and behaviors, allowing for continuous improvement of fact-checking tools.

The User Engagement Scale (UES) is a validated tool designed to measure user engagement across various digital platforms [34]. Its short form, the UES-SF, streamlines user engagement assessment while maintaining reliability, offering a standardized approach for researching engagement with fact-checking tools and linking it to their effectiveness in combating misinformation [44].

While fact-checking initiatives have increasingly been integrated into digital platforms, relatively little synthesis exists on how prior research has measured or explained user engagement with fact-checking tools specifically. Existing studies highlight factors such as trust and usability as key drivers of engagement. For example, a streamlined user interface and rapid response times can enhance user satisfaction [26], while transparency in the fact-checking process builds trust and credibility [25, 45].

Research on user engagement with chatbots within digital fact-checking indicates that specific cultural and relational dynamics further shape these interactions. Specifically, the Auntie Meiyu chatbot, integrated into Taiwan's LINE messaging platform, serves as a critical sociotechnical intervention that mitigates the spread of misinformation by providing automated, real-time fact-checking within private group contexts [10]. However, its

effectiveness in fostering media literacy is often bifurcated by user concerns regarding its robotic delivery, occasional inaccuracies, and inherent privacy risks.

3 METHODOLOGY

This study employs a mixed-methods approach, incorporating both quantitative and qualitative research methods to investigate user engagement with the chatbot of Cek Fakta Tempo. The study utilizes four types of user engagement metrics: a preliminary questionnaire, self-reported questionnaire, semi-structured user interviews, and online engagement metrics. The combination of these methods provides a comprehensive understanding of the audience's interaction and engagement with the chatbot.

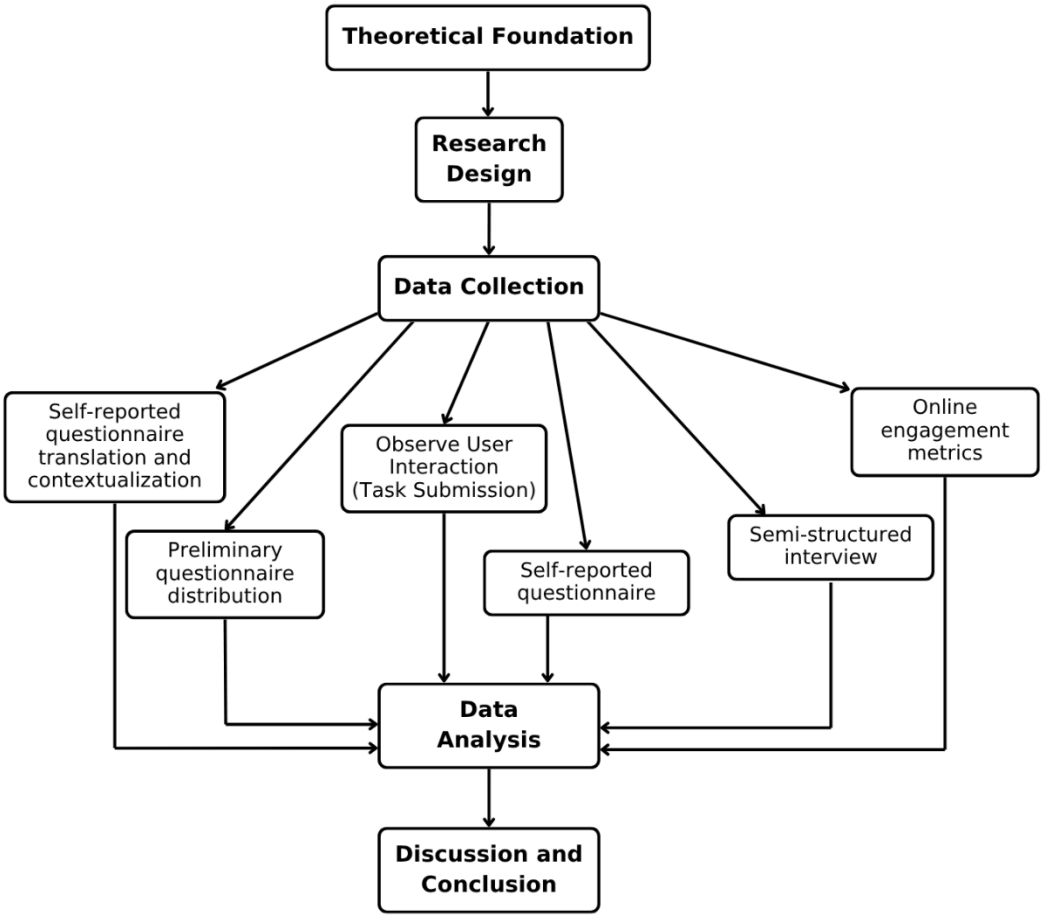


Figure 1: Methodology architecture

3.1 Theoretical Foundation

This study is grounded in the User Engagement Theory [33]. User engagement is the degree to which users find an interactive product appealing and rewarding to use [4, 27]. The degree of engagement depends on the users' perception of the app being rewarding to meet their expectations and determine their future recommendations to other users [5]. In the context of

interventions designed to promote behavioral changes through digital means, user engagement is understood as the degree and nature of interaction with the system, highlighting the importance of both the time spent and the depth of the user's involvement [49].

The User Engagement Theory is used to guide the interpretation of findings and to draw meaningful conclusions about the factors influencing user engagement with the Check Bot of Cek Fakta Tempo and provide valuable lessons for designing user-centric interventions in the fight against misinformation.

3.2 The Case

As one of the IFCN signatories that operates with mainstream media organizations, Cek Fakta Tempo cannot be separated from the Tempo brand, which is synonymous with investigative journalism in Indonesia. Tempo, established in 1971[7], is renowned for its constructive journalism and dedication to combating misinformation. The User Engagement Theory is used to guide the interpretation of findings and to draw meaningful conclusions about the factors influencing user engagement with the Check Bot of Cek Fakta Tempo and provide valuable lessons for designing user-centric interventions in the fight against misinformation.

Tempo magazine was banned in 1982 and again in 1994 for being considered too critical of the New Order regime in Indonesia. After the end of the New Order, the magazine resumed publication on October 6, 1998. The organization later became public and rebranded itself as PT Tempo Inti Media, Tbk., and in 2001 launched its online news portal, Tempointeraktif.com. Its commitment to delivering accurate news and in-depth analyses has solidified its reputation as a trusted source among readers seeking factual reporting.

As an editorial unit launched in 2018, Cek Fakta Tempo actively collaborates with a coalition of fact-checkers across Indonesia. Its journalistic outputs include fact-checking articles, a weekly newsletter, explanatory pieces, and short social media videos. In its efforts, it works closely with Mafindo to develop an integrated hoax database that streamlines joint verification initiatives. This partnership is also reflected in the creation of Cekfakta.com, a collaborative effort between the Indonesian Cyber Media Association (AMSI) and Mafindo.

The chatbot is built on the WhatsApp Business API, utilizing "Check", a premier open-source software platform developed by Meedan. Check[8] empowers Tempo to cultivate knowledge within its communities by facilitating AI-assisted, two-way communication through secure messaging applications. This platform enables the collection of questions and

[7] https://www.tempo.id/corporate.php
[8] https://meedan.org/check

media from the audience, allows for data analysis with Check, and enables the sharing of valuable insights in return. Additionally, the multilingual AI tools help Tempo identify trends and organize related questions, videos, images, and audio files within its Check workspaces.

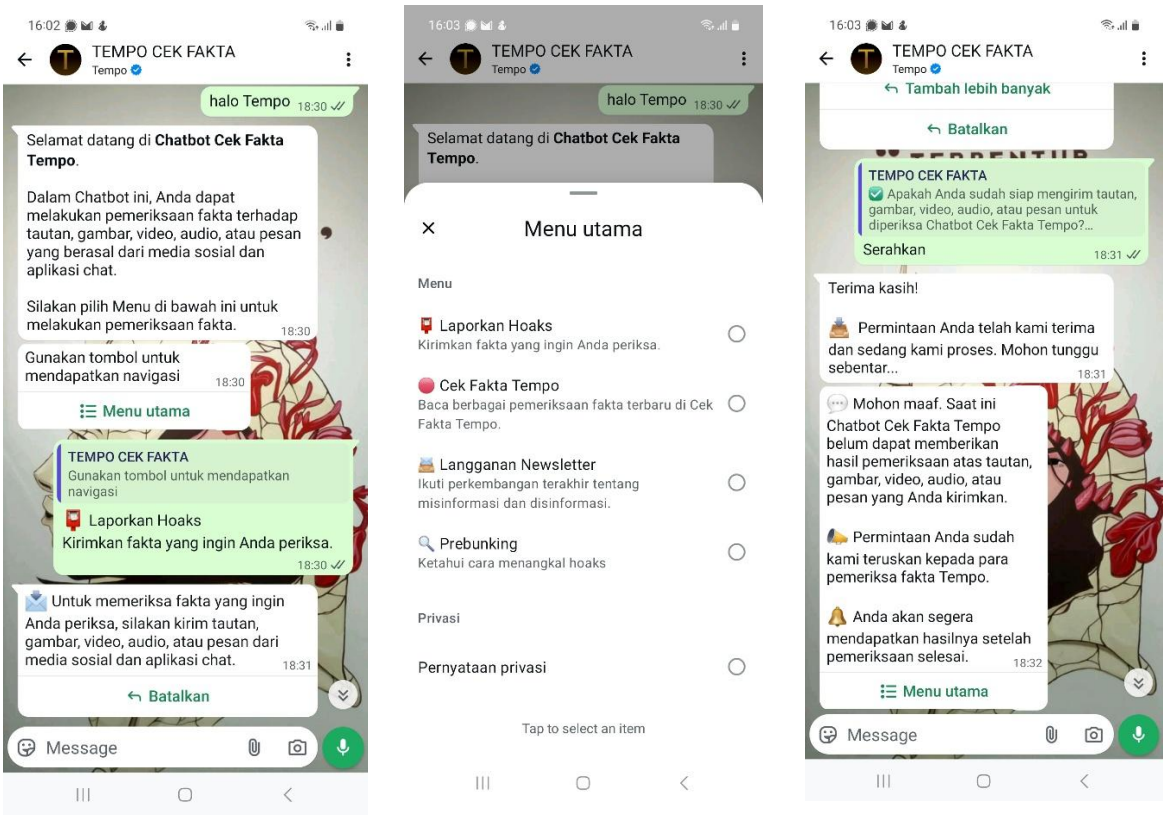


Figure 2: The interface of Tempo's WhatsApp Chatbot Tipline

3.3 The Participants

According to an interview with Fadhli Sofyan, the Head of Audience Development at Tempo, the readership of Tempo's fact-checking channel exhibits distinct demographic characteristics. The predominant age group of readers is under 35 years old, comprising 44.34% of the audience. The second largest age group is 45-55 years old, accounting for 23.33% of the readers. Geographically, the highest concentration of readers comes from cities in Java, including Jakarta, Surabaya, Semarang, and Bandung.

In this study, participants were Tempo readers with practical experience using Tempo's WhatsApp chatbot. This blend of readership habits and direct engagement with digital interfaces provided a comprehensive framework for understanding how participants interact in a digital environment.

3.4 Research Methodology

3.4.1 Preliminary Questionnaire. The researcher employed a preliminary questionnaire to gather data from the audiences of three prominent Indonesian fact-checking organizations:

MAFINDO, Liputan6, and Tempo. The questionnaire was distributed through social media platforms, specifically Telegram and WhatsApp groups. The purpose of this survey was to capture the general experiences and engagement of individuals interacting with the chatbot. The Likert Scale and the 4 subscales of the User Engagement Scale were adapted to measure various aspects of user interaction and engagement.

From February to April 2024, the researcher administered the preliminary questionnaire to the communities of three fact-checking groups that utilize their WhatsApp chatbot, developed by Meedan, a partner of Meta. Additionally, the questionnaire links were disseminated to the WhatsApp group of the Google News Initiative trainers network to reach a broader fact-checking community.

The MAFINDO community's staff member, Muhammad Anshari, assisted in distributing the questionnaire to volunteer WhatsApp groups with a total of about 800 members. The researcher also enlisted the help of Nurfahmi Budiarto, a senior editor at Liputan6, to promote the questionnaire within their WhatsApp groups, aiming to reach an estimated 500 respondents. Furthermore, the researcher disseminated the questionnaire link to the Tempo community via their Telegram channel, which consists of 843 people.

The aim of the preliminary questionnaire was twofold; to identify overarching trends in the engagement with the fact-checking services provided by these organizations and to discern individuals who engage directly with the Cek Fakta Tempo chatbot for follow-up interviews. Of the 14 potential responders that interacted directly with the Cek Fakta Tempo chatbot, only 10 consented to be interviewed.

3.4.2 Submitting tip line and interaction with the chatbot. To observe how participants interact with the WhatsApp chatbot in general, they were asked to submit two tips of contents; one is verified, debunked, and published by Cek Fakta Tempo, one is not verified. The purpose of giving different types of information to tip is for understanding their responses towards different replies from the chatbot.

3.4.3 Self-report questionnaire. The 10 participants were asked to complete a self-report questionnaire aimed to assess objective user engagement, a crucial dimension of user interaction with technology. Self-report measures are widely used in user engagement research because they are “useful for capturing users’ attitudes toward, cognitive appraisals of, and emotions surrounding their experiences of engagement with technology.” The Short Form of the User Engagement Scale (UES) was used to measure key engagement characteristics, providing insights into the participants' perception of their interaction with the chatbot.

3.5 The Data Collection

3.5.1 Semi-structured interviews. Participants were selected based on their responses to the preliminary questionnaire, ensuring that they represent a subset of users who had meaningful interactions with Cek Fakta Tempo's WhatsApp chatbot. Ten respondents who volunteered for semi-structured interviews were chosen from among the 14 respondents who had ever contacted or interacted with it. The interviews were conducted using a flexible set of questions addressing key themes based on the established User Engagement Scale, while also allowing participants to express their experiences freely. During these interviews, a semi-structured format was employed, affording participants ample opportunity to comprehensively recount their individual experiences; precise and targeted prompts facilitated detailed accounts of their motivations for utilizing the chatbot platform, their pre-established expectations pertaining to its operational functionality. All interviews were conducted in Bahasa Indonesia, recorded with the participants' consent, and transcribed verbatim to ensure accuracy and reliability in subsequent analyses.

3.5.2 Online engagement metrics. An analysis was conducted on the WhatsApp chat logs and engagement metrics of the Cek Fakta Tempo chatbot. This examination aimed to uncover patterns in user interaction, including the nature of message exchanges and the types of queries submitted. Data directly related to user engagement was analyzed during the period of the interviews and extended to a year to get broader insights.

4 RESULTS

The study's findings were derived from an in-depth analysis of quantitative and qualitative data. The quantitative data was collected using a preliminary questionnaire to 127 respondents and the short form of the User Engagement Scale (UES-SF) from 10 users of Tempo's WhatsApp fact-checking chatbot. While qualitative feedback was provided through key themes and insights from semi-structured interviews with them. The analysis explores user motivations, experiences, perceptions of the chatbot's design and functionality, and its impact on their decision-making processes.

4.1 Preliminary Questionnaire

Among 127 responses, 101 individuals (79.5%) identified Tempo as a fact-checking organization in Indonesia, while 76 individuals (59.8%) selected Tempo as the most trustworthy media among those with fact check organizations. The majority of respondents (35

individuals or 46.1%) are familiar with Cek Fakta Tempo from the primary news website (Tempo.co), followed by Instagram (26.3%) and referrals from others (23.7%).

Among the 76 persons surveyed, the predominant portion of Fact-Check Tempo's audience (46 individuals or 60.5%) is uninformed about the availability of a WhatsApp chatbot for fact-checking or as a tip line service. The main reasons for their limited involvement with the chatbot are the absence of disinformation to report (54.7%), a lack of comprehension of communication methods (30.7%), and an unclear grasp of its benefits (14.7%).

Of the 30 people, 53% are aware of this chatbot service, but only 14 people, or 46.7 percent, have used it. The majority express satisfaction with the service, emphasizing its simplicity, speed, responsiveness, user-friendliness, interaction, accessibility, and integration. Nonetheless, some consumers expressed dissatisfaction with the service, citing its lack of uniqueness compared to other fact-checking groups and its delayed response time.

4.2 Self-report Questionnaire

4.2.1. Participant Demographics. The study included 10 participants, distributed across four distinct age groups: 10% aged 18–24 years, 40% aged 25–34 years, 40% aged 35–44 years, and 10% aged 45–54 years. Notably, only one participant was female. The majority of participants resided in cities on the island of Java, including Surabaya, Malang, Banyuwangi, and Jember. A smaller number were based in cities outside Java, such as Medan, Buton, and Palu.

4.2.2. Study Task. Each participant was requested to re-engage with the chatbot under controlled conditions to identify behavioral tendencies in response to verified versus unverified information. The author provided two categories of stimulus material: (1) verified content that had undergone Tempo's fact-checking process, published on the Cek Fakta Tempo website, and archived within the tip line system, and (2) unverified misinformation that had not been examined by Tempo. Participants were required to submit both types of content to the chatbot. Verified material would deliver immediate responses to users, while unverified content would offer template replies, necessitating users to await the outcomes of the fact-checking process.

The ten participants were instructed to articulate their actions with each step taken. After completing this assignment, they are required to capture a screenshot of the WhatsApp conversation.

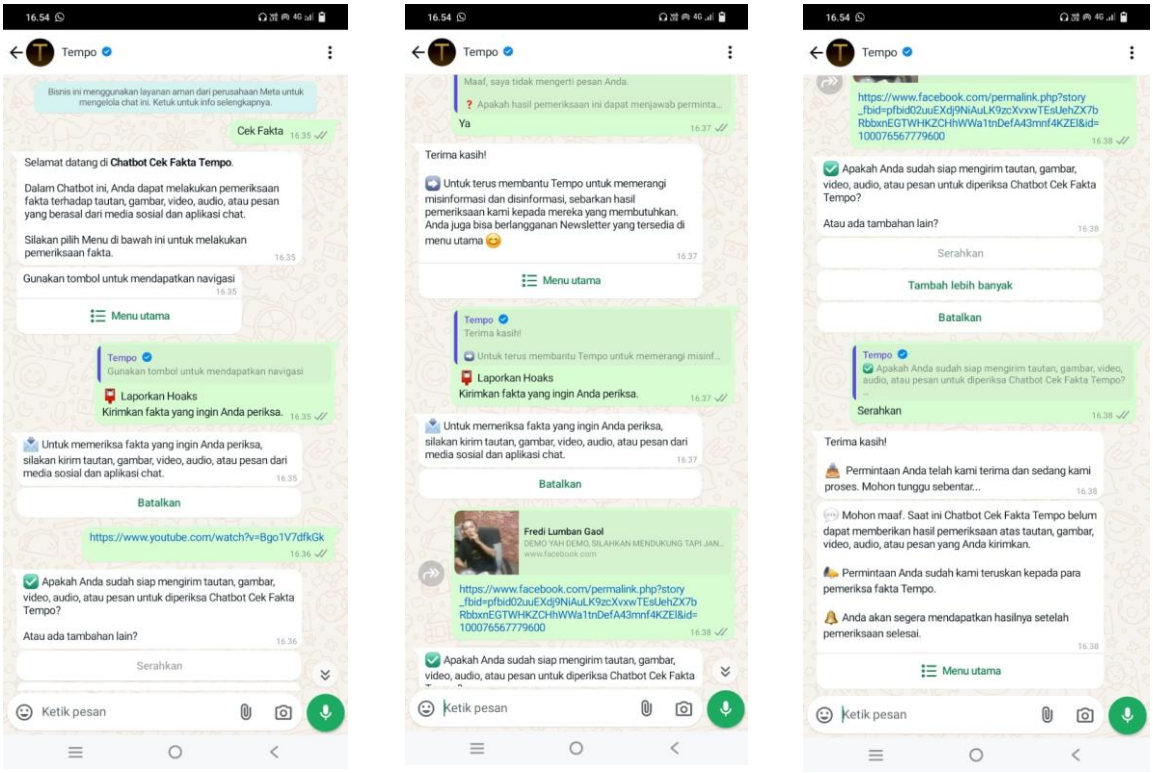


Figure 3: Screenshots from one of the participants

4.2.3. Data Collection and Scoring. After interacting with the chatbot, participants were asked to complete a self-report questionnaire using a 5-point Likert scale. This questionnaire was developed based on the UES-SF framework, consisting of 4 subscales: Focused Attention (FA), Perceived Usability (PU), Aesthetic Appeal (AE), and Reward (RW), and then translated into Indonesian. To analyze the results, the scores for the PU-S1, PU-S2, and PU-S3 subscales were reversed, while the remaining three subscales were computed by summing the response values for the three items in each subscale and then dividing by three.

Table 1: UES Short Form Score Calculation

Participant	FA Score	PU Score	AE Score	RW Score	Overall Engagement Score
1	2.67	5.00	2.00	4.67	3.59
2	3.00	3.33	3.67	5.00	3.75
3	2.33	5.00	3.00	5.00	3.83
4	3.00	4.33	4.33	5.00	4.17
5	2.33	4.00	3.00	4.00	3.33
6	2.33	5.00	3.33	5.00	3.92
7	4.00	4.67	4.33	5.00	4.50
8	3.33	4.00	3.67	4.67	3.92
9	2.33	5.00	5.00	5.00	4.33
10	3.33	5.00	4.67	4.67	4.42

The findings from the User Engagement Scale-Short Form analysis indicate considerable degrees of engagement among individuals utilizing Cek Fakta Tempo's WhatsApp chatbot. The data reveal a nuanced landscape of user interaction, where specific engagement dimensions

elicited distinct responses, ultimately contributing to comprehensive engagement profiles for each participant.

Each participant's answers across all subscales and the overall engagement score show the mean, standard deviation, and range for each subscale.

Table 2: UES Scores and Standard Deviation for All Conditions

Subscale	Mean	Standard Deviation	Range
Focused Attention	2.86	0.592	2.33-4.00 (1.67)
Perceived Usability	4.53	0.593	3.33-5.00 (1.67)
Aesthetic Appeal	3.70	0.909	2.00-5.00 (3.00)
Reward	4.80	0.321	4.00-5.00 (1.00)
Overall Engagement	3.97	0.378	3.33-4.50 (1.17)

The Focused Attention subscale had the lowest average score, suggesting users did not fully immerse themselves while using the chatbot. This aligns with qualitative feedback, where users described their interactions as brief and focused on specific tasks. Many noted they used the chatbot alongside other activities, indicating it was efficient but not highly engaging. Rather than engaging users emotionally or cognitively, it serves as a rapid-response utility. This mirrors O'Brien and Toms'[34] finding that low immersive engagement is typical in tools designed for brief transactional use.

The Perceived Usability subscale yielded a high score, underscoring the chatbot's robust interface design and smooth interaction flow. Participants uniformly described the chatbot as intuitive, responsive, and user-friendly, largely attributed to its integration within WhatsApp, a platform already embedded in their daily communication routines. The chatbot's low entry barriers and simplified command structure rendered it accessible even for users with limited digital proficiency. This finding is particularly significant in a context like Indonesia, where mobile-first access and platform familiarity are key to digital inclusion [29].

The Aesthetic Appeal dimension exhibited a moderate level of scoring, with some variability across participants. While not a primary driver of initial engagement, aesthetic elements such as emoji usage and an uncluttered menu design contributed to a sense of ease and comfort. Some participants noted that these features made the experience feel more personalized and enjoyable. Although this dimension did not significantly influence users' decision to engage with the chatbot, it likely impacted their level of comfort in using it, particularly over repeated interactions.

The Reward Factor subscale yielded the highest score, corroborating the finding that the instrumental and social outcomes of chatbot usage are central to user engagement. Participants

consistently highlighted the satisfaction they derived from verifying claims, assisting others, and sharing credible information. These benefits encompassed both cognitive and social rewards.

4.2.4 Descriptive Statistics. This section shows distributional differences across four subscales, illustrated by boxplots within violin plots in Fig. 4. They combine density estimation with traditional box plot elements to provide a comprehensive view of data distribution patterns. Each violin's width represents the Kernel Density Estimate (KDE) at different score levels, with wider sections indicating higher concentrations of responses.

The embedded box plots display the median (central line), interquartile range (box boundaries), and whiskers extending to the data range, offering both distributional shape and summary statistics simultaneously.

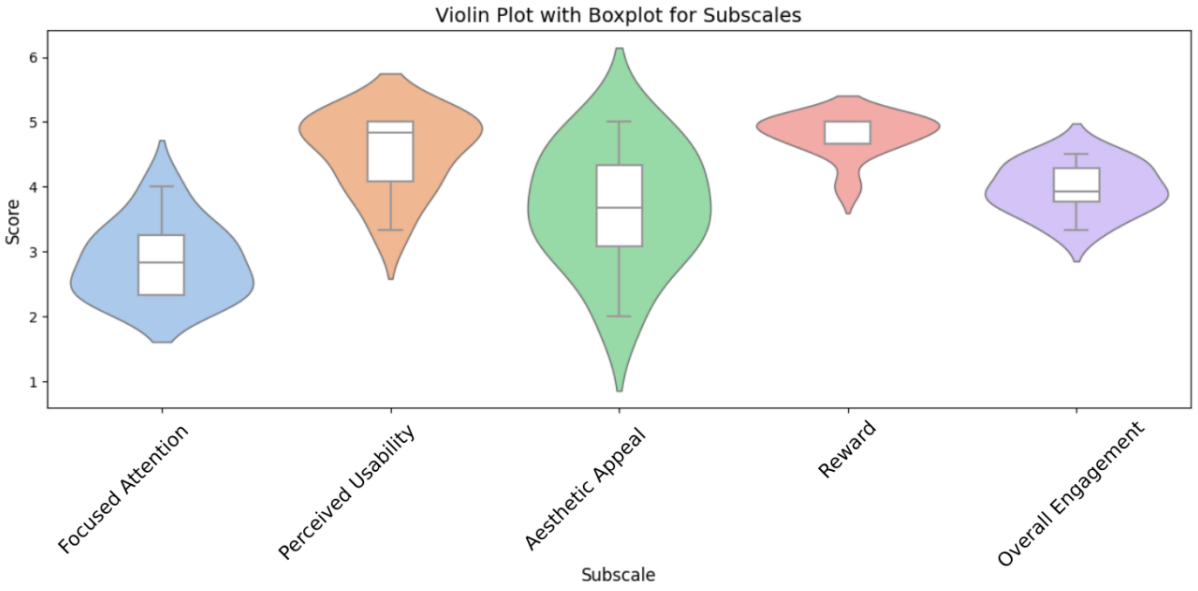


Figure 4: UES Result in a Boxplot inside Violin Plot

Analysis of the subscale distributions reveals considerable variability in response patterns across the five dimensions. The Aesthetic Appeal subscale demonstrates the broadest distribution, with scores ranging from approximately 1.5 to 6 and a relatively uniform density across this range, suggesting substantial heterogeneity in participant perceptions. Similarly, Perceived Usability exhibits a broad distribution spanning scores from 3 to 5.5, though with greater concentration in the upper range. In contrast, Focused Attention shows a narrower, more symmetric distribution centered around 2.5-3.5, indicating more consistent responses among participants for this dimension.

Notably, the Reward subscale presents a distinctly concentrated distribution, with scores tightly clustered between 4.5 and 5 and minimal variance, reflecting strong consensus among

respondents regarding this aspect of the assessment. Overall Engagement shows moderate variability, with scores primarily distributed between 3.5 and 5, centered at approximately 4.

4.3 Thematic Analysis

The researcher utilized open coding and thematic analysis to identify major themes, supported by detailed descriptive codes, that collectively reveal the cognitive, practical, and social dynamics underlying user engagement with the chatbot. The themes were derived using an inductive and iterative approach, allowing patterns to emerge from the data rather than being pre-defined.

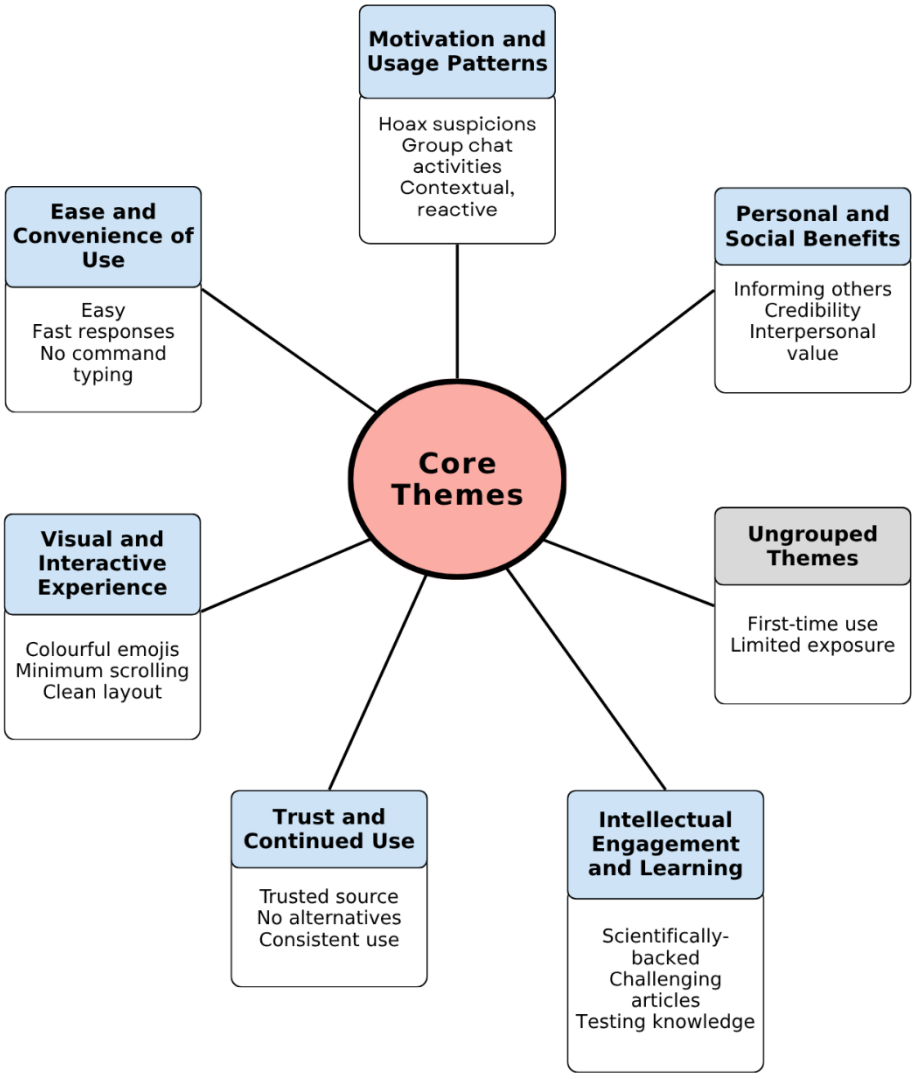


Figure 5: Thematic Analysis major themes

The findings are structured around six central themes, each capturing distinct yet interconnected aspects of user incentives and experiences. The analysis also identifies frequently occurring individual codes, which are supplemented by rich participant quotations in the original Indonesian language, accompanied by English translations.

4.3.1 Ease and Convenience of Use. This theme was cited by all ten participants, indicating that ease of use is a universal factor in engaging with the chatbot. Users emphasized that the integration with WhatsApp lowered barriers to entry. Key characteristics include minimal typing requirements, fast system responsiveness, lightweight interface, and intuitive menu navigation.

"Langsung dikasih pilihan. Tidak perlu mengetik lagi." (I was immediately given the options. No need to type anything.) (P1)

"Tinggal menyapa, lalu kirim. Tidak ribet." (Just greet, then send. Not complicated.) (P8)

"Sangat mudah." (Very easy.) (P9)

These responses demonstrate that technical simplicity and platform familiarity significantly enhance user willingness to engage.

4.3.2 Motivation and Usage Patterns. Ninety percent of participants described their engagement with the chatbot as situational rather than habitual. The chatbot was predominantly used in response to specific incidents—especially within social group chats—where participants encountered suspicious or provocative information.

"Kalau saya ragu, saya bakal cek fakta lewat chatbot ini." (If I'm doubtful, I will check the facts through this chatbot.) (P10)

"Kalau dapat info hoax di grup WhatsApp keluarga, aku langsung kirimkan nomor chatbot." (If I get hoax info in the family WhatsApp group, I immediately send them the chatbot number.) (P9)

Such usage patterns reflect a reactive model of fact-checking behavior, wherein engagement is driven by immediate social or informational needs rather than proactive inquiry.

4.3.3 Personal and Social Benefits. Eight out of ten participants reported that chatbot engagement provided interpersonal value, especially in their roles as informal fact-checkers within their social networks. The ability to quickly verify and then share accurate information with family or peer groups reinforced users' sense of responsibility and credibility.

"Untuk menghindari perdebatan dalam kehidupan sosial kita, salah satu caranya adalah kita memberikan info dengan data jadi." (One way to avoid debates in our social lives is to provide information with factual data.) (P1)

"Saya pake hasil cek fakta Tempo untuk meluruskan informasi yang berseliweran." (I use Tempo's fact-check results to clarify misinformation that's circulating.) (P8)

These findings suggest that social empowerment, trust-building, and reputational gains within one's digital community constitute key incentives for continued use.

4.3.4 Visual and Interactive Experience. This theme emerged from seven participants who commented on the aesthetic design and user interface of the chatbot. Although not always the primary motivator, elements such as emoji integration, compact message design, and minimal scrolling contributed to user satisfaction and repeated engagement.

> *"Misalnya ada pertanyaan belum dicek, saya ingat ada gambar lonceng dan chatbot mendeskripsikan "kami akan memberitahukan anda ketika sudah selesai dicek". buatku menarik."* (For example, if there are unchecked questions, I remember there was a bell icon and the chatbot described, 'We will notify you when it has been checked.' I find that interesting.) (P4)

> *"Emoji-emoji menjadikan Cek Fakta Tempo lebih berwarna dan lebih asik."* (Emojis make Tempo's fact-checking interface more colorful and enjoyable.) (P10)

Thus, while not decisive in initiating engagement, the design choices enhance retention and usability over time.

4.3.5 Trust and Continued Use. Seven participants highlighted their trust in Tempo as a credible news organization as a central reason for ongoing engagement. The chatbot's institutional affiliation offered an additional layer of legitimacy that alternative services may not possess.

> *"Kalau sudah ditulis Tempo, Tempo jadi parameter. Jadi kalau kebenarannya sudah diuji oleh Tempo, sudah jadi jaminan."* (Once it has been written by Tempo, Tempo becomes the benchmark. So if the truth has been verified by Tempo, it is guaranteed.) (P8)

> *"Tempo selalu menjadi andalan untuk mengecek isu-isu nasional."* (Tempo is always the go-to for checking national issues.) (P9)

> *"Iya, saya akan tetap menggunakan untuk mengecek tulisan cek fakta terbaru."* (Yes, I will continue to use it to check the latest fact-checking articles.) (P10)

Such trust not only supports continued use but also amplifies user confidence in sharing verified content publicly or socially. Trust plays a pivotal role in the adoption and sustained use of technology, particularly in the context of information services [1].

4.3.6 Intellectual Engagement and Learning. Half of the participants noted some form of cognitive stimulation stemming from their use of the chatbot, particularly through reading linked fact-check articles. The interaction itself was not seen as intellectually demanding, but the content accessed through the bot contributed to learning and critical thinking.

> *"Kita jadi tahu cek fakta basisnya ilmiah karena benar-benar dilakukan pengecekan."* (We can see that fact checking is scientifically-based because it is actually checked) (P8)

This suggests that the chatbot serves as a gateway to verified information, reinforcing media literacy and critical evaluation skills.

4.4 Online Engagement Metrics

This data was retrieved from Cek Fakta Tempo's Check Bot dashboard provided by Meedan. Extracted metrics are the number of WhatsApp conversations, service conversations, unique users, returning users, positive searches, and negative searches. Amongst all types of data, service conversations before January 2024 were not available.

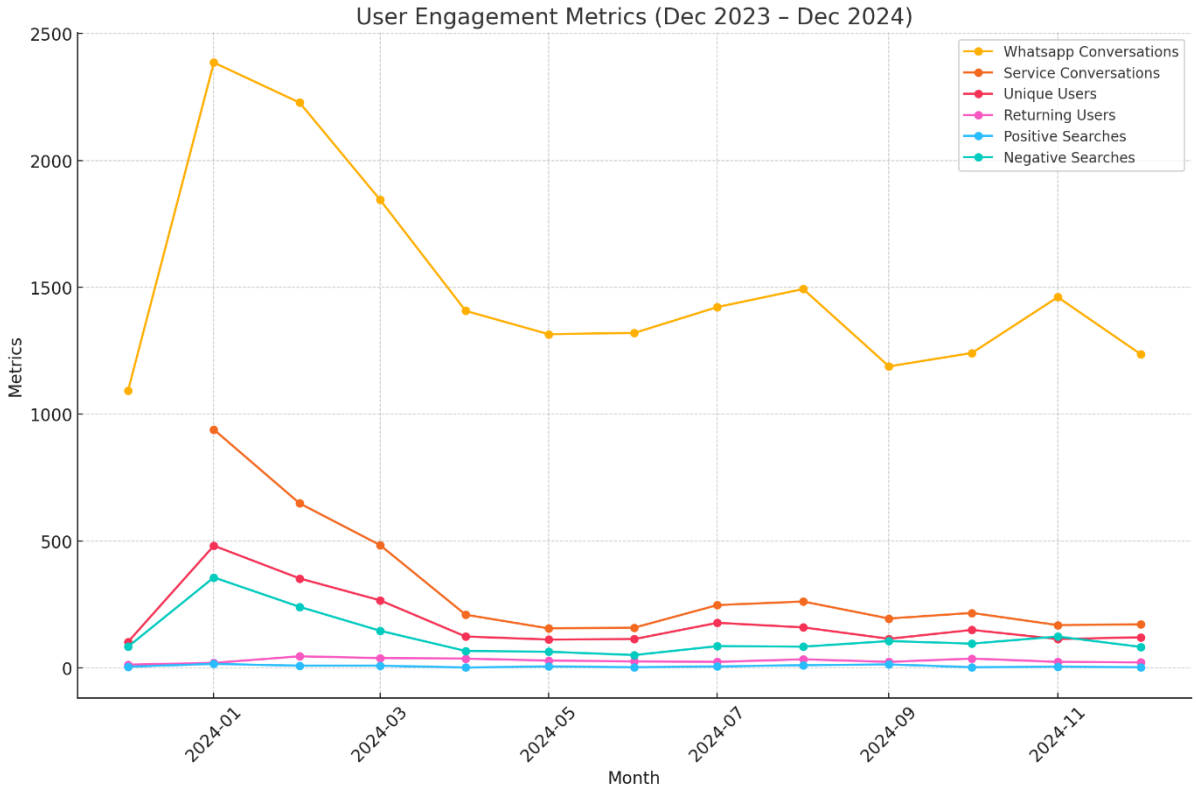


Figure 6: Online Metrics provided by Check Bot, Meedan

As illustrated in Fig. 6, the total number of conversations on WhatsApp shows a variable pattern throughout the 13-month observation period. The levels of engagement display several spikes, especially in the early and mid-2024 timeframe, which align with periods of increased public activity. A notable rise was observed in February 2024, coinciding with the Presidential Election on February 14, 2024, which fueled significant political discussion and media attention. Another spike was seen in November 2024, corresponding with the Simultaneous Regional Elections on November 27, 2024. It is likely that both events influenced the surge in engagement, as users sought immediate updates, clarifications, and interactions through the WhatsApp platform.

The volume of service-related conversations generally mirrors the overall WhatsApp activity trend, though on a smaller scale. Data from Meedan since January 2024 indicates that

the metrics for both unique and returning users follow a consistent trend. Both measures show a gradual increase during the first half of 2024, followed by slight fluctuations in the subsequent months.

Positive search results remained relatively stable but low throughout the analytical period, indicating that a portion of users consistently accessed relevant information or content. In contrast, negative search results occurred more often and showed significant variability. These findings point to potential challenges regarding content discoverability or the relevance ranking within the search algorithm. However, months with a higher incidence of negative searches frequently aligned with greater overall engagement, suggesting that increased activity may also lead to a rise in unsuccessful search attempts.

5 DISCUSSION

Using a combination of thematic analysis from in-depth interviews and quantitative data from the User Engagement Scale – Short Form (UES-SF), this study sought to uncover the incentives, behaviors, and challenges associated with chatbot-mediated fact-checking. The discussion is structured around the two research questions in relation to existing literature on digital literacy, misinformation, and chatbot-based interventions. This chapter will also examine how this case study contributes to broader understandings of fact-checking behavior in private messaging environments.

5.1 User Motivations and Engagement Patterns

The primary motivation for users to engage with the Cek Fakta Tempo chatbot stems from its instrumental utility and contextual relevance. Participants reported using the chatbot in response to encountering questionable content, particularly within social contexts such as WhatsApp group discussions. Engagement was largely reactive, driven by a need to confirm or refute information circulating in their networks. This aligns with previous research highlighting how misinformation often spreads in semi-private, peer-oriented environments [51], prompting users to seek verification when their own credibility or social responsibility is at stake.

Participants emphasized the convenience of the platform, with the majority highlighting the ease of use afforded by WhatsApp integration. The chatbot's placement within a familiar messaging interface allowed users to access fact-checking tools without the need to download a separate app or navigate complex menus. The menu-driven design, minimal input requirements, and rapid response times further contributed to its appeal. In particular, the chatbot’s capacity to respond quickly and its menu-based options were frequently mentioned

as significant facilitators of engagement. These attributes correspond to what O'Brien and Toms [34] describe as conditions that support "transparent interaction," in which users experience minimal disruption in completing their intended task.

Beyond individual verification needs, participants frequently referenced a desire to share accurate information with others as a key motivator. Many described scenarios in which they used the chatbot to obtain verified information that could then be relayed to others, typically family members or community contacts. In doing so, participants positioned themselves as informal gatekeepers or mediators of credible content. This behavior supports the notion of socially situated fact-checking [47], wherein the drive to fact-check is not only cognitive but relational, embedded in the user's social ecosystem.

Additionally, trust in Tempo's institutional credibility also emerged as a powerful incentive. Multiple participants expressed that the reason they chose to use the Cek Fakta Tempo chatbot—as opposed to similar services—was due to Tempo's longstanding reputation as a neutral and reputable news source. This finding underscores the importance of source credibility in shaping digital engagement [31], especially in contexts where institutional trust is a scarce commodity. Participants perceived that information verified by Tempo held greater legitimacy, making it easier for them to defend or circulate in conversations.

Aesthetic features, while not commonly cited as primary motivators, contributed to an overall positive user experience. Elements such as emojis, a clean visual layout, and limited text complexity were appreciated for reducing the perceived burden of fact-checking. Though these features did not drive initial use, they likely played a role in maintaining engagement over time by making the interaction more emotionally accessible.

Taken together, these findings indicate that engagement with the chatbot is rooted in a combination of pragmatic, social, and trust-based incentives. The tool functions less as a source of continuous interaction and more as a responsive mechanism, a fact-checking utility embedded into everyday communication flows.

5.2 The Affordances and Challenges That Influence User Engagement of a Fact-Checking Chatbot

Even though the User Engagement Scale–Short Form (UES-SF) offers a valuable framework for assessing user engagement by focusing on usability, aesthetics, and reward, a more complete understanding necessitates the inclusion of each metrics. Based on the results of the UES-SF, the Focused Attention subscale, which yielded the lowest mean value, indicates that participants' interactions were brief and transactional in nature. It is further reinforced by the Aesthetic Appeal subscale, which demonstrated the largest standard deviation, suggesting

substantial variation in participants' visual preferences. To bridge the gap between these two dimensions, the use of WhatsApp emoticon templates that align with Indonesian cultural contexts may be employed. As noted by Stark and Crawford [52], emojis function not only to maintain social relations within the logic of economic instrumentalism and efficiency but also serve as phatic expressions in digital communication. Similarly, the use of language perceived as neutral in AI-mediated interactions can be effectively applied to facilitate interpersonal communication around sensitive topics—particularly in WhatsApp groups that encompass multiple generations—where disagreement is more likely to occur [10].

However, institutional trust is not captured within the four standard UES-SF dimensions while repeatedly being emphasized in the qualitative data. Several participants noted that Tempo's brand credibility was a decisive factor in their willingness to trust and use the chatbot. This underscores the importance of source reputation in evaluating the perceived value of information, which is particularly crucial in the context of fact-checking [12].

Based on online engagement metrics, a small but consistent group of users continued engaging with the chatbot, likely because their expectations were met or the system aligned with their information-seeking routines. This group represents valuable informants for understanding successful retention, as the chatbot's features, such as aesthetic appeal and rewards, were compatible with their goals.

The consistent growth of both unique and returning users throughout 2024 suggests that the platform successfully supports user acquisition while maintaining retention. The synchronization of user engagement with national events, specifically the Presidential Election on February 14, 2024, and the Simultaneous Regional Elections on November 27, 2024, demonstrates the platform's responsiveness to the sociopolitical environment. The strong correlation between WhatsApp conversations, service interactions, and user retention emphasizes the interdependent nature of engagement within digital communication systems. Such findings underscore the critical role of accessible, real-time fact-checking tools in fostering informed public discourse, particularly within high-stakes political contexts where misinformation can proliferate rapidly [30].

Despite overall positive engagement trends, the persistently high number of negative searches reveals limitations in the platform's information retrieval mechanisms. This issue underscores the need for improving content discoverability and query accuracy, which are central to enhancing user experience and reducing informational friction. From a theoretical perspective, this aligns with the uses and gratifications framework, [48] as users actively seek

content that satisfies specific informational needs. When unmet, engagement quality declines despite high activity volume.

The prevalence of negative searches also points to the necessity of refining chatbot algorithms to better interpret nuanced user queries and provide more precise, relevant responses, thereby enhancing the perception of usefulness and reliability [43]. Furthermore, enhancing these aspects can significantly influence continuous usage intention, as observed in studies where information quality and system quality positively affect user satisfaction and retention [3, 11].

Therefore, optimizing content organization, metadata tagging, and search logic could significantly improve user satisfaction and strengthen the relationship between engagement intensity and informational fulfillment. While user engagement is influenced by contextual events and system responsiveness, its long-term sustainability depends on continuous improvement in content accessibility and user-centered design [38].

5.3 Method Criticism and Limitations

Despite the valuable insights gained, this study had some limitations. The findings may lack broader applicability due to its limited participant pool of ten individuals on interview. It may also affect the UES-SF quantitative insights, so the small sample size restricts the ability to draw robust statistical conclusions.

The results are shaped by the specific Indonesian media and messaging landscape, necessitating cautious generalization to other contexts. Furthermore, the limited geographical and linguistic scope of the study participants restricts the generalizability of these findings to other populations and cultures, requiring further investigation.

Additionally, the inherent subjectivity of self-reported engagement metrics, as captured by the UES-SF, presents a limitation, given that these measures may not always align with objective behavioral data. Moreover, relying solely on self-reported data might overlook subconscious or implicit user behaviors that contribute to engagement but are not explicitly articulated by participants.

5.4 Future Work

Future research could administer the UES-SF to a larger, more diverse sample to study engagement patterns. Comparing Tempo's chatbot to similar offerings could reveal interface or content differences influencing user behavior. Tracking users' chatbot engagement over time could shed light on fact-checking habits. Investigating whether explanations of the fact-checking process increase trust or engagement would be valuable.

Future studies could investigate the potential relationship between users' trust in fact-checkers and their corresponding behaviors by incorporating a larger, more diverse participant group and extend the analysis to various geopolitical and linguistic settings to validate the observed patterns. Additionally, the existing UES-SF framework could be expanded to incorporate a fifth dimension: trust in the information source.

Investigating the interplay between chatbot design elements, such as anthropomorphism and communication styles, influence emotional responses could provide deeper insight into user retention. Future work could also examine how large language model (LLM)-based chatbots enhance personalization, conversational depth, and user trust in fact-checking interactions.

5.5 Ethical and Societal Consequences

The deployment of fact-checking chatbots, particularly in politically charged environments such as Indonesia, carries significant ethical and societal implications. On a positive note, such platforms play a crucial role in fostering informed public discourse by providing accessible, real-time fact-checking tools that combat the rapid proliferation of misinformation during high-stakes political contexts. By enhancing information quality and accessibility, these tools can empower citizens to make more informed decisions, thereby strengthening democratic processes. The ability of tools like Tempo's chatbot to address social and informational needs contributes to a more responsible and socially conscious communication landscape.

Beyond the limited references concerning user engagement in a WhatsApp chatbot, a further ethical challenge arises from the divergent patterns of usage of the WhatsApp messaging application in Indonesia. The incorporation of fact-checking tiplines within chatbot interfaces introduces inherent data management and privacy risks, particularly within conversational ecosystems characterized by their private nature. Entities deploying these services, such as fact-checking organizations and media outlets, may inadvertently or intentionally facilitate the collection of personal identifiers, including telephone numbers and platform-specific usernames, alongside user behavioral data.

If implemented within European Union member states, these data management practices could entail considerable legal and ethical consequences, particularly concerning GDPR compliance. In Indonesia, however, this issue has not yet garnered substantial attention.

6 CONCLUSION

This study provides empirical evidence and user-centric insights into the dynamics of misinformation, fact-checking, and technology adoption in Indonesia, with broader

implications for the Global South. It contributes to the emerging literature exploring how users engage with fact-checking tools in private messaging environments. The findings indicate that user engagement with the Cek Fakta Tempo WhatsApp chatbot is primarily driven by ease of use, trust, and the ability to address social and informational needs.

Although users do not deeply interact with the interface itself, the chatbot plays a crucial role in facilitating informed, responsible, and socially conscious communication. As misinformation continues to evolve, the tools that counter it must also adapt. Well-designed, trusted, and promoted chatbots like Tempo's represent a scalable and effective mechanism for defending truth in everyday digital life.

ACKNOWLEDGMENTS

This master's thesis and the opportunity to pursue my graduate studies were made possible by the Indonesia Endowment Fund for Education (LPDP), supported by the contributions of the Indonesian people through their taxes. I am deeply grateful for this support, which paved the way for my research. My profound thanks go to my supervisor, Björn Thuresson, for his invaluable guidance, time, and unwavering support throughout this project. I also wish to thank my examiner, Tino Weinkauf; as well as the dedicated educators Leif Dahlberg, Daniel Pargman, Elina Eriksson, Mark Comber, Mario Romero Vega; and Erik Ceder Göransson, the EECS master coordinator, whose commitment made the Media Management program such a rich and illuminating experience. The realization of this endeavor was made possible by their collective patience and invaluable contributions.

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A APPENDIXES

A.1 Preliminary Questionnaire

Q1. Which fact-checking organizations/institutions/media do you perceive? (you can choose more than one)

Tempo, Mafindo, Liputan6.com, Cekfakta.com, Kompas.com, Tirto.id, The Conversation Indonesia, JawaPos.com, AFP, Kominfo, other else

Q2. Of the organizations/institutions/media that have Fact Check channels, which ones are the most trusted so you access/read the most often? (select one)

Tempo, Liputan6, Mafindo

Q3. How did you first get to know Tempo Fact Check?

Website, Instagram, Recommended by other people, or WhatsApp

Q4. Which Tempo Fact Check's platforms do you perceive?

Main website, Instagram, YouTube, Facebook, or TikTok

Q5. Did you know that Tempo has a WhatsApp number in the form of a chatbot that you can contact to ask questions about hoaxes?

Yes or No

Q6. Have you ever contacted or had interaction with the Fact Check chatbot?

Yes or No

Q7. What do you think are the advantages of Tempo's WhatsApp chatbot compared to other fact-checking organisations or media?

Q8. In your opinion, how attractive is the visual appearance of the Tempo Fact Check's article page?

Very interesting, Interesting, Quite interesting, Less attractive, Very uninteresting

Q9. In your opinion, how attractive is the visual appearance of the Tempo Fact Check's video?

Very interesting, Interesting, Quite interesting, Less attractive, Very uninteresting

Q10. What is your opinion towards the Tempo Fact Check's articles?

It's very easy to understand, It's easy to understand, It's a bit confusing, It's difficult to understand, It's very difficult to understand

Q11. Tempo Fact Check is the best at checking facts amidst the onslaught of hoaxes

Very agree, Agree, Not sure, Disagree

Q12. Tempo Fact Check is the most qualified fact checker

Very agree, Agree, Not sure, Disagree

Q13. How willing are you to recommend or share Check Fact Tempo contents with others?

I'm very willing, I'm willing, Not sure, I'm not willing

Q14. How rewarding is Tempo Fact Check's content for you?

Very rewarding, Rewarding, Not sure, Not rewarding

Q15. Why haven't you used/interacted with Tempo's Fact Check chatbot? (for respondents who answered "No" on Q6)

I do not have any hoaxes to report/check, I do not understand how to communicate with the chatbot, or I do not know its benefits

A.2 User Engagement Scale-Short Form Adaptation

Original English Version	Adaptation	Indonesian Version
FA-S.1 I lost myself in this experience. FA-S.2 The time I spent using Application X just slipped away. FA-S.3 I was absorbed in this experience. PU-S.1 I felt frustrated while using this Application X . PU-S.2 I found this Application X confusing to use. PU-S.3 Using this Application X was taxing. AE-S.1 This Application X was attractive. AE-S.2 This Application X was aesthetically appealing. AE-S.3 This Application X appealed to my senses. RW-S.1 Using Application X was worthwhile. RW-S.2 My experience was rewarding. RW-S.3 I felt interested in this experience.	FA-S.1 I lost myself in this experience. FA-S.2 The time I spent using Tempo Fact Check WhatsApp chatbot just slipped away. FA-S.3 I was absorbed in this experience. PU-S.1 I felt frustrated while using Tempo Fact Check WhatsApp chatbot . PU-S.2 I found Tempo Fact Check WhatsApp chatbot confusing to use. PU-S.3 Using Tempo Fact Check WhatsApp chatbot was taxing. AE-S.1 Tempo Fact Check WhatsApp chatbot was attractive. AE-S.2 Tempo Fact Check WhatsApp chatbot was aesthetically appealing. AE-S.3 Tempo Fact Check WhatsApp chatbot appealed to my senses. RW-S.1 Using Tempo Fact Check WhatsApp chatbot was worthwhile. RW-S.2 My experience was rewarding. RW-S.3 I felt interested in this experience.	FA-S.1 Saya lupa diri saat menggunakan chatbot ini FA-S.2 Waktu yang saya habiskan untuk menggunakan WA Cek Fakta Tempo terasa berlalu begitu saja. FA-S.3 Saya asyik dengan pengalaman ini. PU-S.1 Saya merasa frustasi saat menggunakan WA Cek Fakta Tempo ini. PU-S.2 Saya merasa WA Cek Fakta Tempo ini membingungkan untuk digunakan. PU-S.3 Menggunakan WA Cek Fakta Tempo ini menyusahkan. AE-S.1 WA Cek Fakta Tempo ini menarik. AE-S.2 Secara desain estetik, WA Cek Fakta Tempo ini menarik. AE-S.3 WA Cek Fakta Tempo ini menarik bagi indra saya. RW-S.1 Menggunakan WA Cek Fakta Tempo ini bermanfaat. RW-S.2 Pengalaman menggunakan WA chatbot ini bermanfaat. RW-S.3 Saya merasa aktivitas Cek Fakta dengan WA Tempo ini menarik.

A.3 Semi-structured Interview Questions

In Indonesian

Q1. Ceritakan pengalaman secara umum Anda menggunakan WhatsApp chatbot untuk memeriksa fakta?

Q2. Seberapa sering Anda merasa benar-benar tenggelam/fokus saat menggunakan chatbot fact-check di WhatsApp?

Q3. Seberapa mudah atau sulit menggunakan chatbot Cek Fakta Tempo di WhatsApp?

Q4. Apakah ada fitur atau aspek tertentu dari chatbot yang menurut Anda sangat ramah pengguna atau bahkan menantang?

Q5. Apa pendapat Anda tentang desain visual dan tata letak chatbot fact-check?

Q6. Bagaimana desain chatbot mempengaruhi keinginan Anda untuk menggunakannya?

Q7. Aspek apa saja dari penggunaan chatbot CF Tempo yang menurut Anda menyenangkan?

Q8. Apakah Anda merasa penggunaan chatbot merangsang atau menantang secara intelektual?

Q9. Seberapa berguna chatbot dalam kehidupan sehari-hari atau proses pengambilan keputusan Anda?

Q10. Apakah Anda menerima umpan balik (*feedback*) atau pengakuan atas penggunaan chatbot, mungkin dari rekan, kolega, anggota keluarga Anda?

Q11. Seberapa besar kemungkinan Anda akan terus menggunakan chatbot pemeriksa fakta di masa mendatang?

Q12. Faktor apa yang berkontribusi pada keputusan Anda untuk terus menggunakan atau berhenti menggunakan chatbot?

Q3. Peningkatan atau perubahan apa yang akan Anda sarankan untuk meningkatkan keterlibatan Anda dengan chatbot?

In English

Q1. Describe your general experience using WhatsApp chatbots to check facts.

Q2. How often do you feel completely immersed/focused when using fact-check chatbots on WhatsApp?

Q3. How easy or difficult is it to use Tempo's Fact Check chatbot on WhatsApp?

Q4. Are there any features or aspects of the chatbot that you find particularly user-friendly or even challenging?

Q5. What is your opinion of the visual design and layout of the fact-check chatbot?

Q6. How does the chatbot's design influence your desire to use it?

Q7. What aspects of using the Tempo Fact Check chatbot do you find enjoyable?

Q8. Do you find using the chatbot stimulating or intellectually challenging?

Q9. How useful is the chatbot in your daily life or decision-making process?

Q10. Do you receive feedback or recognition for using the chatbot, perhaps from your peers, colleagues, or family members?

Q11. How likely are you to continue using the fact-checking chatbot in the future?

Q12. What factors contribute to your decision to continue using or stop using chatbots?

Q13. What improvements or changes would you suggest to increase your engagement with chatbots?

TRITA-EECS-EX-2026:66
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